

Indian Institute of Technology Hyderabad

Matrix Theory (AI1000): Homework-1

Date: 4th September, 2026

1. Suppose all vectors x in the unit square $0 \leq x_1 \leq 1, 0 \leq x_2 \leq 1$ are transformed to Ax (A is 2×2).
- (a) What is the shape of the transformed region (all Ax)?
 - (b) For which matrices A is that region a square?
 - (c) For which A is it a line?
 - (d) For which A is the new area still 1?

2. In the vector space P_3 of all $p(x) = a_0 + a_1x + a_2x^2 + a_3x^3$, let S be the subset of polynomials with $\int_0^1 p(x) dx = 0$. Verify that S is a subspace and find a basis.

3. Suppose

$$T(M) = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} M \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}.$$

Find a matrix with $T(M) \neq 0$. Describe all matrices with $T(M) = 0$ (the kernel of T) and all output matrices $T(M)$ (the range of T).

4. The transformation T that transposes every matrix is definitely linear. Which of these extra properties are true?

- (a) $T^2 = \text{Identity Transformation}$
- (b) The Kernel of T is the zero matrix
- (c) Every matrix is in the range of T .
- (d) $T(M) = -M$ is impossible.

5. If $AB = 0$, the columns of B are in the nullspace of A . If those vectors are in $\mathbb{R}^n$, prove that

$$\text{rank}(A) + \text{rank}(B) \leq n.$$

6. Suppose A is the sum of two matrices of rank one:

$$A = uv^T + wz^T.$$

- (a) Which vectors span the column space of A ?
- (b) Which vectors span the row space of A ?
- (c) The rank is less than 2 if _____ or if _____.
- (d) Compute A and its rank if $u = z = (1, 0, 0)$ and $v = w = (0, 0, 1)$.

7. Describe the four subspaces of $\mathbb{R}^3$ associated with

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \quad \text{and} \quad I + A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}.$$

8. Construct a matrix with the required property, or explain why you can't.

- (a) Column space contains $\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$, $\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$, row space contains $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$, $\begin{bmatrix} 2 \\ 5 \end{bmatrix}$.
- (b) Column space has basis $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$, nullspace has basis $\begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}$.
- (c) Dimension of nullspace = 1 + dimension of left nullspace.
- (d) Left nullspace contains $\begin{bmatrix} 1 \\ 3 \end{bmatrix}$, row space contains $\begin{bmatrix} 3 \\ 1 \end{bmatrix}$.
- (e) Row space = column space, nullspace $\neq$ left nullspace.

9. Suppose A is 5×4 with rank 4. Show that $Ax = b$ has no solution when the 5×5 matrix $[A \ b]$ is invertible. Show that $Ax = b$ is solvable when $[A \ b]$ is singular.

10. If A is a 64×17 matrix of rank 11, how many independent vectors satisfy $Ax = 0$? How many independent vectors satisfy $A^T y = 0$?

11. Find a counterexample to the following statement: If v_1, v_2, v_3, v_4 is a basis for the vector space $\mathbb{R}^4$, and if W is a subspace, then some strict subset of the v 's is a basis for W .

12. What are the special solutions to $Rx = 0$ and $R^T y = 0$ for these R ?

$$R = \begin{bmatrix} 1 & 0 & 2 & 3 \\ 0 & 1 & 4 & 5 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad R = \begin{bmatrix} 0 & 1 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

13. If A has rank r , then it has an $r \times r$ submatrix S that is invertible. Find that submatrix S from the pivot rows and pivot columns of each A :

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 2 & 4 \end{bmatrix} \quad A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{bmatrix} \quad A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

14. Explain why all these statements are false:

- (a) The complete solution is any linear combination of x_p and x_n .
- (b) A system $Ax = b$ has at most one particular solution.
- (c) The solution x_p with all free variables zero is the shortest solution (minimum length $\|x\|$). (Find a 2×2 counterexample.)
- (d) If A is invertible there is no solution x_n in the nullspace.

15. Write the complete solutions $x = x_p + x_n$ to these systems:

$$\begin{bmatrix} 1 & 2 & 2 \\ 2 & 4 & 5 \end{bmatrix} \begin{bmatrix} u \\ v \\ w \end{bmatrix} = \begin{bmatrix} 1 \\ 4 \end{bmatrix} \quad \begin{bmatrix} 1 & 2 & 2 \\ 2 & 4 & 4 \end{bmatrix} \begin{bmatrix} u \\ v \\ w \end{bmatrix} = \begin{bmatrix} 1 \\ 4 \end{bmatrix}.$$

16. Which of the following are subspaces of $\mathbb{R}^\infty$?

- (a) All sequences like $(1, 0, 1, 0, \dots)$ that include infinitely many zeros.
- (b) All sequences $(x_1, x_2, \dots)$ with $x_j = 0$ from some point onward.
- (c) All decreasing sequences: $x_{j+1} \leq x_j$ for each j .
- (d) All convergent sequences: the x_j have a limit as $j \rightarrow \infty$.
- (e) All arithmetic progressions: $x_{j+1} - x_j$ is the same for all j .
- (f) All geometric progressions $(x_1, kx_1, k^2x_1, \dots)$ allowing all k and x_1 .

17. Which of the following descriptions are correct? The solutions x of

$$Ax = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

form

- (a) a plane.
 - (b) a line.
 - (c) a point.
 - (d) a subspace.
 - (e) the nullspace of A .
 - (f) the column space of A .
18. (a) Describe a subspace of M that contains $A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ but not $B = \begin{bmatrix} 0 & 0 \\ 0 & -1 \end{bmatrix}$.
- (b) If a subspace of M contains A and B , must it contain I ?
- (c) Describe a subspace of M that contains no nonzero diagonal matrices.
19. Suppose P is a plane through $(0, 0, 0)$ and L is a line through $(0, 0, 0)$. The smallest vector space containing both P and L is either _____ or _____.
20. To decide whether b is in the subspace spanned by $w_1, \dots, w_n$, let the vectors w be the columns of A and try to solve $Ax = b$. What is the result for
- (a) $w_1 = (1, 1, 0)$, $w_2 = (2, 2, 1)$, $w_3 = (0, 0, 2)$, $b = (3, 4, 5)$?
 - (b) $w_1 = (1, 2, 0)$, $w_2 = (2, 5, 0)$, $w_3 = (0, 0, 2)$, $w_4 = (0, 0, 0)$, and any b ?